Definition 10.3

A function f is said to be **differentiable in the closed interval** $[a,b]$ if it is differentiable on the open interval (a,b) and at the end points a and b,

$$f'(a) = \lim_{\Delta x \to 0^+} \frac{f(a+\Delta x)-f(a)}{\Delta x} = \lim_{h \to 0} \frac{f(a+h)-f(a)}{h}, \quad h>0$$

$$f'(b) = \lim_{\Delta x \to 0^-} \frac{f(b+\Delta x)-f(b)}{\Delta x} = \lim_{h \to 0} \frac{f(b-h)-f(b)}{-h}, \quad h>0.$$

If f is differentiable at $x = x_0$, then $f'(x_0) = \lim_{x \to x_0} \frac{f(x)-f(x_0)}{x-x_0}$, where $x = x_0 + \Delta x$ and $\Delta x \to 0$ is equivalent to $x \to x_0$. This alternative form is some times more convenient to be used in computations.

As a matter of convenience, if we let $h = \Delta x$, then $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$, provided the limit exists.

10.3 Differentiability and Continuity

Illustration 10.3

Test the differentiability of the function $f(x) = |x - 2|$ at $x = 2$.

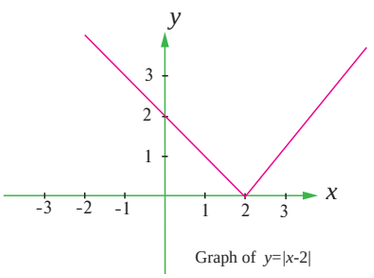


Fig. 10.18

Solution

We know that this function is continuous at $x = 2$.

But
$$f'(2^-) = \lim_{x \to 2^-} \frac{f(x)-f(2)}{x-2}$$
$$= \lim_{x \to 2^-} \frac{|x-2|-0}{x-2}$$
$$= \lim_{x \to 2^-} \frac{|x-2|}{x-2} = \lim_{x \to 2^-} \frac{-(x-2)}{(x-2)} = -1 \text{ and}$$

$$f'(2^+) = \lim_{x \to 2^+} \frac{f(x)-f(2)}{x-2}$$
$$= \lim_{x \to 2^+} \frac{|x-2|-0}{x-2} = \lim_{x \to 2^+} \frac{(x-2)}{(x-2)} = 1$$

Since the one sided derivatives $f'(2^-)$ and $f'(2^+)$ are not equal, $f'(2)$ does not exist. That is, f is not differentiable at $x = 2$. At all other points, the function is differentiable.

If $x_0 \neq 2$ is any other point then

$$f'(x_0) = \lim_{x \to x_0} \frac{|x-x_0|}{x-x_0} = \begin{cases} 1 & \text{if } x > x_0 \\ -1 & \text{if } x < x_0 \end{cases}$$

Thus
$$f'(2) = \begin{cases} 1 & \text{if } x > 2 \\ -1 & \text{if } x < 2 \end{cases}$$

The fact that $f'(2)$ does not exist is reflected geometrically in the fact that the curve $y = |x - 2|$ does not have a tangent line at (2, 0). Note that the curve has a sharp edge at (2, 0).

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